

Chapter 3 Machine Learning Workflows

Austin R.J. Downey

Overfitting

Training and testing on the same dataset is a major mistake.

- ▶ The model can memorize the training data
- ▶ Training accuracy may appear perfect
- ▶ Performance on new data may be poor

This problem is called:

Overfitting

Goal:

- ▶ Build models that generalize to unseen data

Training and Testing Sets

Standard supervised learning workflow:

- ▶ Train model using training data
- ▶ Evaluate model using test data
- ▶ Test data must remain unseen during training

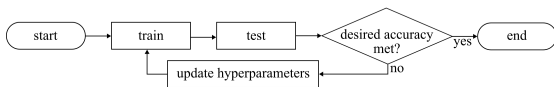
Notation:

$\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}}$

$\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}}$



(a)



(b)

Training/testing split.

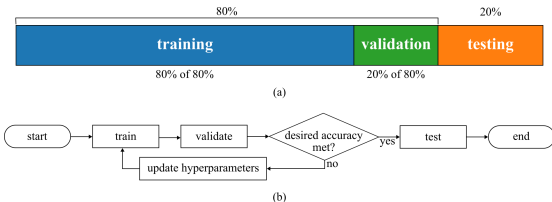
Validation Sets

A validation set is often added to tune models.

- ▶ Training set:
 - ▶ Learn model parameters
- ▶ Validation set:
 - ▶ Tune hyperparameters
- ▶ Test set:
 - ▶ Final evaluation

Scikit-Learn:

```
train_test_split
```



Training, validation, and testing workflow.

Machine Learning Pipelines

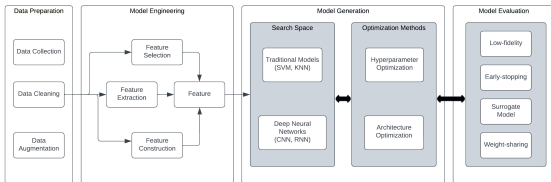
Scikit-Learn pipelines combine multiple processing steps into a single workflow.

Typical steps:

- ▶ Data preprocessing
- ▶ Feature scaling
- ▶ Feature engineering
- ▶ Model training

Benefits:

- ▶ Cleaner code
- ▶ Consistent preprocessing
- ▶ Easier cross-validation
- ▶ Simplified hyperparameter tuning



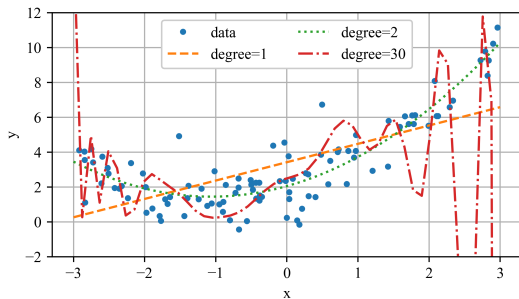
Example machine learning pipeline.

Learning Curves

Model complexity strongly affects performance.

- ▶ Low complexity:
 - ▶ Underfitting
- ▶ High complexity:
 - ▶ Overfitting
- ▶ Intermediate complexity:
 - ▶ Better generalization

Learning curves help diagnose model behavior.



Underfitting, good fit, and overfitting.

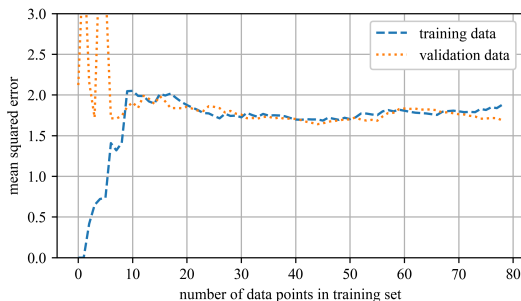
Underfitting

Characteristics of underfitting:

- ▶ Model is too simple
- ▶ Cannot capture data complexity
- ▶ Training and validation errors remain high

Learning curves:

- ▶ Curves are close together
- ▶ Errors plateau at high values



Learning curves for an underfitting model.

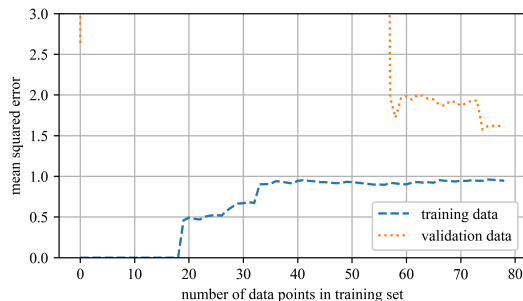
Overfitting

Characteristics of overfitting:

- ▶ Model is too complex
- ▶ Fits noise in the training data
- ▶ Excellent training performance
- ▶ Poor validation performance

Learning curves:

- ▶ Large gap between curves
- ▶ High variance



Learning curves for an overfitting model.

Well-Fit Model

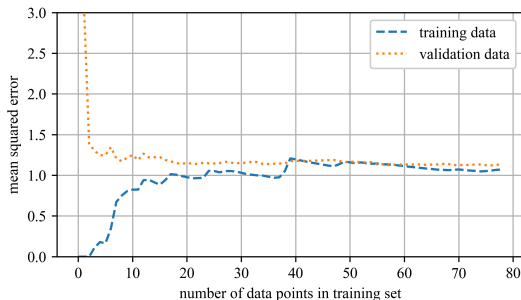
A good model balances:

- ▶ Bias
- ▶ Variance

Characteristics:

- ▶ Low training error
- ▶ Low validation error
- ▶ Small gap between curves

The model generalizes well to unseen data.



Learning curves for a well-fit model.

Interpreting Learning Curves

Learning curves compare:

Training Error vs. Validation Error

- ▶ Underfitting:
 - ▶ Both errors high
 - ▶ Curves close together
- ▶ Overfitting:
 - ▶ Training error low
 - ▶ Validation error much higher
- ▶ Good fit:
 - ▶ Both errors low
 - ▶ Curves converge

Example 2.3: Learning Curves

- ▶ Compare learning curves for:
 - ▶ Linear regression
 - ▶ Polynomial regression
- ▶ Plot training and validation errors as the training set grows
- ▶ Demonstrates:
 - ▶ Underfitting
 - ▶ Overfitting
 - ▶ Model generalization
 - ▶ Bias–variance tradeoff
- ▶ Shows how model complexity affects performance

Regularized Linear Models

Overfitting can be reduced through:

Regularization

Idea:

- ▶ Constrain model complexity
- ▶ Prevent fitting noise in the data
- ▶ Improve generalization

For polynomial models:

- ▶ Use a lower polynomial degree

For linear models:

- ▶ Penalize large coefficient values

Ridge Regression

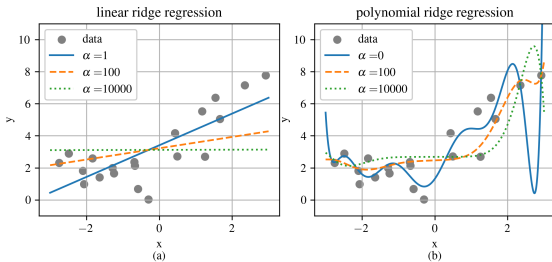
Ridge Regression adds a penalty term to the cost function:

$$J(\theta) = \text{MSE}(\theta) + \alpha \frac{1}{2} \sum_{i=1}^n \theta_i^2$$

- ▶ α : regularization strength
- ▶ Large $\alpha \rightarrow$ smaller weights
- ▶ Reduces overfitting

Note

The bias term θ_0 is not regularized.



Ridge regression with different α values.

Example 2.4: Ridge Regression

- ▶ Apply Ridge Regression to a high-degree polynomial model
- ▶ Add regularization to reduce overfitting
- ▶ Use a pipeline with:
 - ▶ `PolynomialFeatures`
 - ▶ `StandardScaler`
 - ▶ `Ridge`
- ▶ Demonstrates:
 - ▶ Smoother predictions
 - ▶ Reduced variance
 - ▶ Improved model stability
- ▶ Compare different values of α

Early Stopping

Early stopping is a form of regularization for iterative learning methods.

Idea:

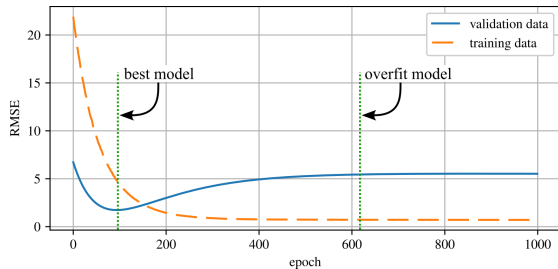
- ▶ Monitor validation error during training
- ▶ Stop training when validation error begins increasing

Benefits:

- ▶ Prevents overfitting
- ▶ Improves generalization
- ▶ Simple and computationally inexpensive

Key Idea

Stop training at the minimum validation error.



Early stopping prevents overfitting.

Example 2.5: Early Stopping

- ▶ Train a high-degree polynomial model using:
 - ▶ Stochastic Gradient Descent
- ▶ Track:
 - ▶ Training error
 - ▶ Validation error
- ▶ Demonstrates:
 - ▶ How overfitting develops during training
 - ▶ How validation error identifies the optimal stopping point
 - ▶ Improved generalization through early stopping
- ▶ Training halts once validation performance begins degrading